

Smart Irrigation Advisory & Recommendation Engine

Mathematical Formulas, Decision Logic & Dataset Sources Documentation

1. Multi-Factor Agronomic Scoring Formula

For any candidate irrigation practice T in {Drip, Sprinkler, AWD, Furrow, Border, Raised Bed, Pitcher, Flood}, the total agronomic suitability score Score(T) is calculated as:

$Score(T) = 0.35 * S1(T) + 0.20 * S2(T) + 0.20 * S3(T) + 0.15 * S4(T) + 0.10 * S5(T)$

Factor Weighting Breakdown:		
Weight	Agronomic Factor	Description & Impact
35%	S1: Groundwater Stage	Aquifer stress (Safe, Semi-Critical, Critical, Over-Exploited)
20%	S2: Soil Texture & Infiltration	Soil water-holding capacity (Coarse, Medium, Fine)
20%	S3: Crop Water Demand	Seasonal evapotranspiration demand (Kc x ET0)
15%	S4: Rainfall & Weather	Observed & forecast precipitation shortfall
10%	S5: Practice Transition	Preference penalty/bonus vs existing farm practice

2. Calibrated AI Confidence Score Formula

Confidence percentage reflects decision certainty based on top-pick score alignment, consensus factor agreement, and runner-up margin separation:

$Confidence\ (%) = MIN(96, MAX(72, (Score_top * 0.90) + (N_agree / 5 * 12) + MIN(0.8 * Margin, 12)))$

3. Dynamic Water Volume Saved Formula

Water savings per hectare are calculated dynamically based on published crop seasonal water requirements (ETc):

$Volume\ Saved\ (m3/ha) = Crop\ Water\ Requirement\ (m3/ha) * (WaterSavingsPct / 100)$

Crop Water Savings Benchmarks:

Crop Species	Seasonal Req (m3/ha)	Drip Saved (55%)	Sprinkler Saved (35%)
Paddy / Rice	12,500 m3/ha	6,875 m3/ha	4,375 m3/ha
Sugarcane	18,000 m3/ha	9,900 m3/ha	6,300 m3/ha
Cotton	7,000 m3/ha	3,850 m3/ha	2,450 m3/ha
Wheat	4,500 m3/ha	2,475 m3/ha	1,575 m3/ha
Potato / Tomato	5,500 - 6,000 m3/ha	3,025 - 3,300 m3/ha	1,925 - 2,100 m3/ha
Mustard / Bajra	3,000 m3/ha	1,650 m3/ha	1,050 m3/ha

4. Official Dataset Sources & Scientific Citations

1. Survey of India (SOI)
vb_so_i_hr.kmz — Official 7,010 Village Boundary Polygons for Haryana.
2. Central Ground Water Board (CGWB)
Dynamic Groundwater Assessment Reports (1991-2020 & 2023-2024) — Net availability, draft,

extraction stage %, DTW.

**3. Open-Meteo / ECMWF
ERA5 & GFS**

Historical & forecast
precipitation, temp, humidity,
Penman-Monteith ET0.

4. FAO-56 (United Nations)

Crop Evapotranspiration
guidelines, Kc factors,
application efficiency (Drip
90%, Sprinkler 75%, Flood
40%).

5. ICAR & CCS HAU Hissar

Package of Practices for
Haryana Crops, soil texture
maps, AWD paddy protocols.

6. NASA SMAP & Sentinel-1

0-15cm Topsoil volumetric
moisture dataset for Haryana
districts (2018-2025).